A Network Analysis of Candidate–Party Dynamics in Pakistan's National Assembly Elections (2008, 2013, and 2024)

Introduction:

Our network focuses on understanding the political structure of Pakistan by constructing a bipartite network where candidates and political parties serve as nodes across the 2008, 2013, and 2024 National Assembly elections, using data from the Gallup Pakistan Elections database (1970-2024). From a Bipartite graph, it moves towards a unipartite graph: party-to-party network to understand how candidates switch take place among parties, apply network, centrality metrics, and community detection algorithms to understand the structures, connectivity, and stability within these networks, helping us reflect on fragmentation and coherence of the overall political structure.

Bipartite Graph:

In our bipartite graph, an edge is established between a candidate node and a party node if the candidate contested an election in any of these years under that party's banner. In this network, an edge is weighted by the number of unique election years (2008, 2013, or 2024) a candidate contested with a specific party. A weight of 1 means the candidate contested with the party in only one of the election years. A weight of 2 or 3 indicates a candidate's repeated association and loyalty to a party across multiple elections. Another key aspect of our network is that it focuses on candidates who switch parties, so during node creation, it identifies candidates who have an edge connected to more than one unique party node. There are 196 unique political parties/affiliations and 9,876 candidates. The total edges are 10,746 unique candidates' party connections.

Unipartite Graph: **Party to Party Network**:

Metric	Value
Number of Parties (Nodes)	**196**
Number of Connections (Edges)	**390**
Network Density	**0.0204 (2.04%)**
Average Path Length	**2.82 steps**
Network Diameter	**7 steps**
Global Clustering Coefficient	**0.232**
Average Local Clustering Coefficient	**0.759**
Number of Connected Components	**84**
Largest Component Size	**113 parties**
Largest Component Percentage	**57.7%**

Figure 1: Global Network metrics summary table

In this network, as shown in the above summary table, there are 196 party nodes and 390 edges. The edges are defined by when one candidate moves from one party to another; an edge exists between two parties. The edge shows how candidates flow between parties over the period of election cycles, which is important in understanding the dynamics of party loyalty and defection. This is an undirected network. The network density is 0.0204 (2.04%), which indicates that relative to the total number of unique political parties, the candidates switching only connects a very small fraction of possible pairs. This further highlights how the network is sparse and existence of fragmentation, where there are many connections between parties possible, which are not yet explored. In this context, it also means limited connectivity between parties in terms of candidate switching.

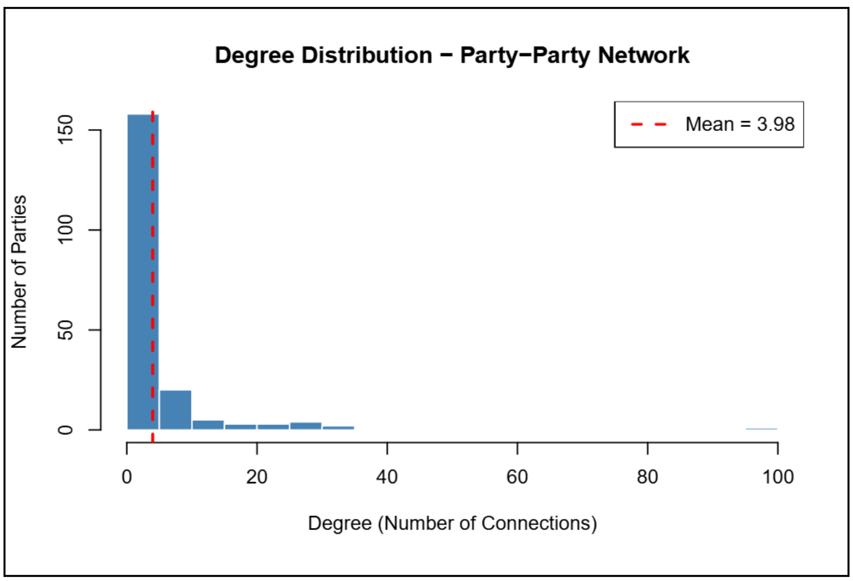

Figure 2: Degree distribution (histogram)

The average degree is 3.98, which shows that, on average, each party has been involved in candidate switching with approximately 4 other unique parties. Contextualising this in the number of parties (196), it reinforces how most parties are not part of the candidate switching process. The above degree distribution through histogram also shows that the distribution is highly skewed, where most political parties have low degrees, while a few actually dominate candidate switching. This indicates that candidate stitching is not a central democratic activity within Pakistan's structure, rather concentrated within a small core.

The number of components shows that the network is highly fragmented with 84 disconnected subgraphs, and the largest component size is 113 parties (57.7%), showing that less than 60% of parties belong to the main interconnected bloc, while the remaining 42.3% are scattered in small isolated groups. This shows that in the largest component party, switching actually occurs; however, small, isolated groups either have low political mobility or are very constrained to their own area, with limited interaction with the main parties.

The Average path length is 2.82, showing that, on average, it takes fewer than 3 candidates to move from one political party to another. It highlighted, in terms of party switching, that the communication spreads very quickly within the largest connected component. This also means that Pakistan’s political system over the three electoral cycles has evolved into a highly efficient system of candidate switching, which also means exchanging political influence and sentiment, despite a sparse network; connections that exist are strategically placed, creating an efficient path. This can also indicate that there is feasibility for rapid shifts in alliances or changes in political loyalty, which can lead to political volatility. The network diameter, the longest shortest path between two political parties, is 7 candidates, thus showing that while an efficient reach exists, there are peripheral parties within the main component that increase the diameter.

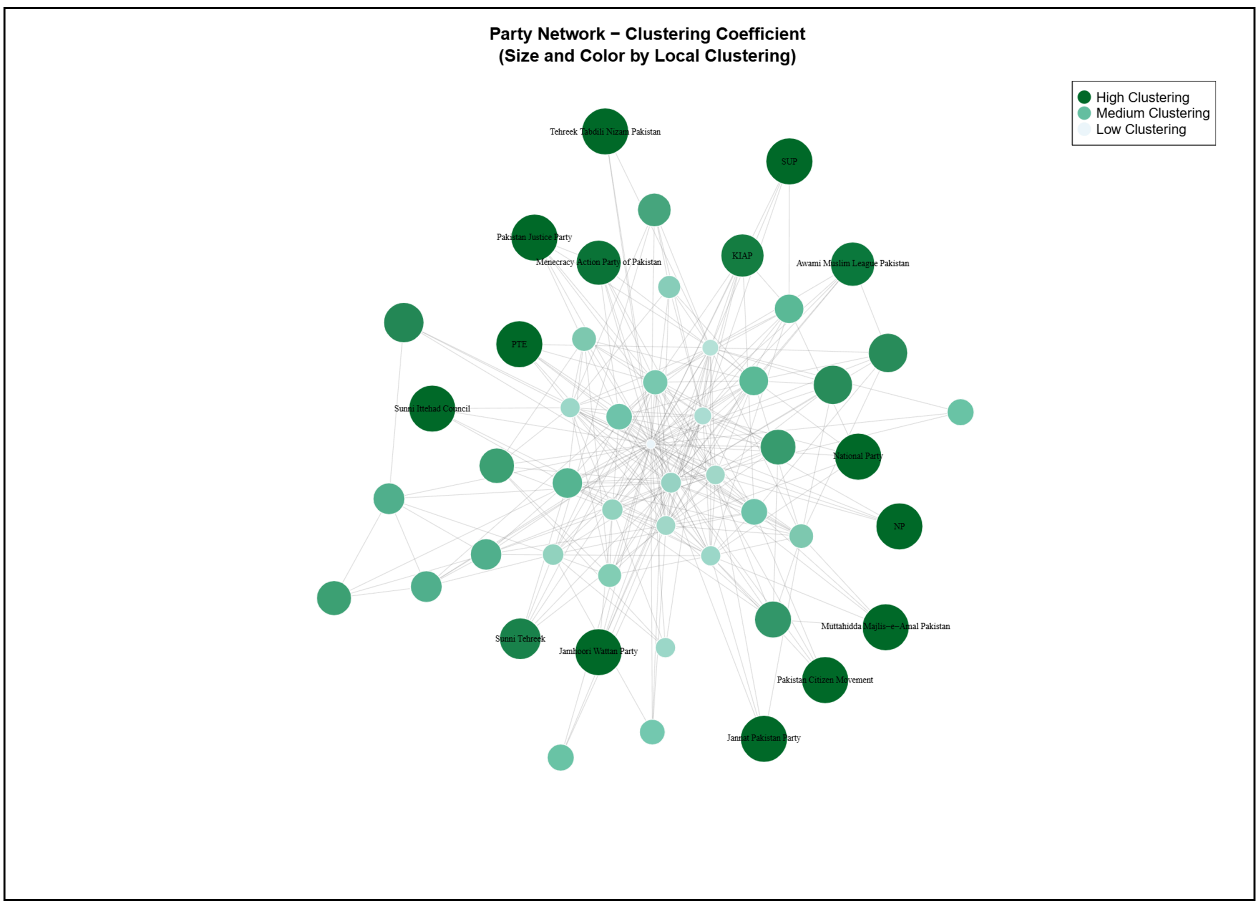

Figure 3: Local Clustering Coefficient

The average local clustering coefficient is 0.759, which shows that there is high local cohesion where parties tend to form tight-knit communities or alliances with their neighbours through candidate switching. This shows that the overall network may be sparse, but many individual parties maintain strong connections through candidate switching. They share common candidates, which might also mean similar regional areas or policies, thus forming closely knit subgroups as shown in dark green and large size and labels of top nodes with the highest clustering coefficient. Large Dark Green nodes form highly clustered cliques. Parties like the Pakistan Justice Party and many more have a more tightly knit group in terms of candidates switching to their nearest neighbour. Regional Parties also have a high clustering coefficient, which reflects that within their regional politics, they are likely to form close-knit groups based on candidates switching to their nearest regional party. This shows that they need not be central in position as part of mainstream politics; rather, candidate switching also takes place in regional areas over the period of three election cycles.

Centrality Measures:

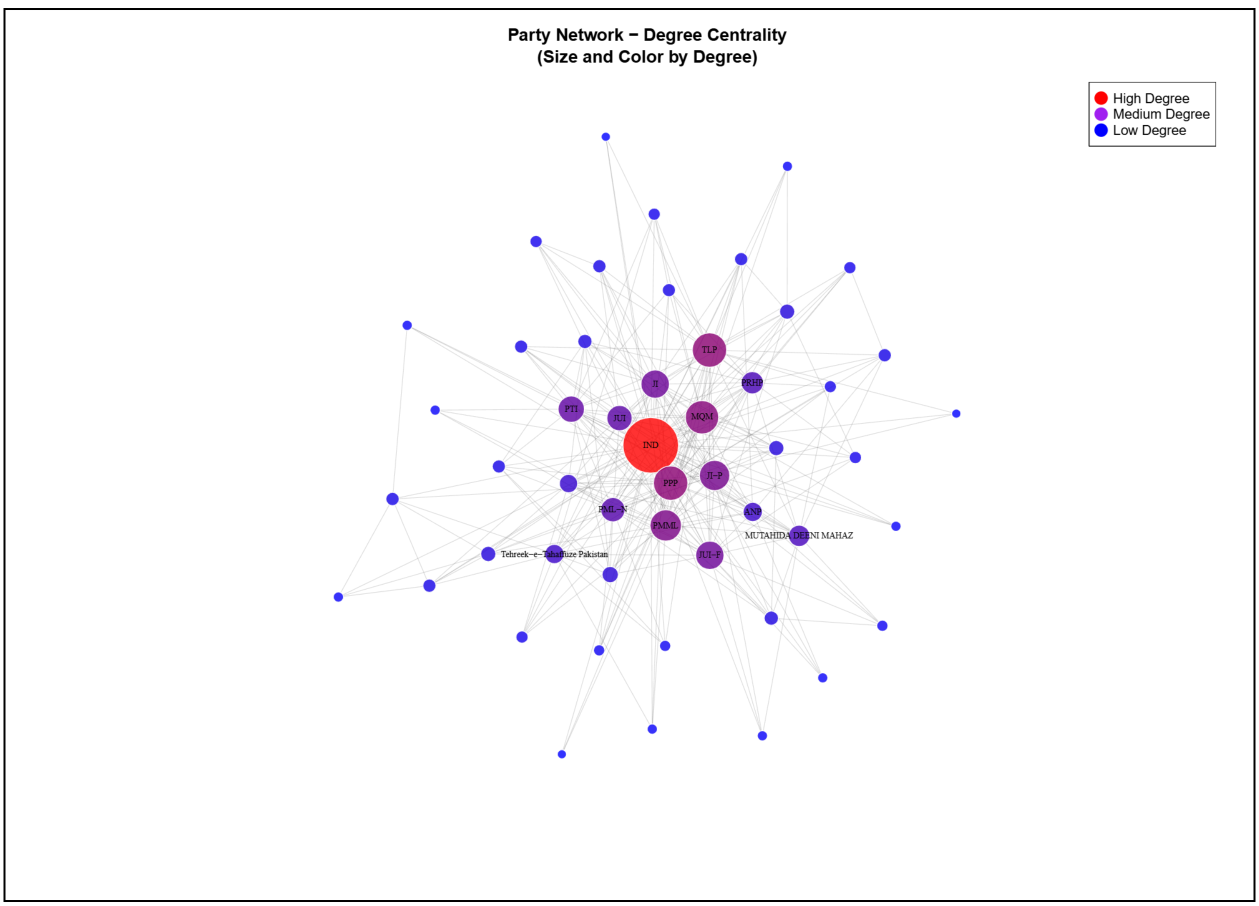

Figure 4: Degree Centrality

The degree centrality shows that IND (independent) candidates have the highest number of degrees, as shown by the Red colour and largest in size. This makes IND (independent) a central hub connecting it to half of all parties. This shows that independent candidates' affiliation is most accessible, and it means it is a source or destination for candidate switching over the three periods, thus providing better exposure. This reflects a pattern in Pakistan where candidates from major political parties switch to run as independents. Independents also act as a neutral way for candidates who want to avoid affiliation or who lose tickets in large parties. This high degree for an independent candidate can indicate instability in the party allegiance. Followed by IND, the second most central nodes are PPP, MQM, and TLP, which are among the largest political parties in Pakistan, observed from three periods. Their high degree shows that they have a consistent role in candidate switching in three election periods, where they frequently acquire or lose candidates. This suggests a strategic position. The blue colored and small nodes indicate parties with a very low degree. They are rarely involved as part of candidates switching between them, showing their role as more peripheral.

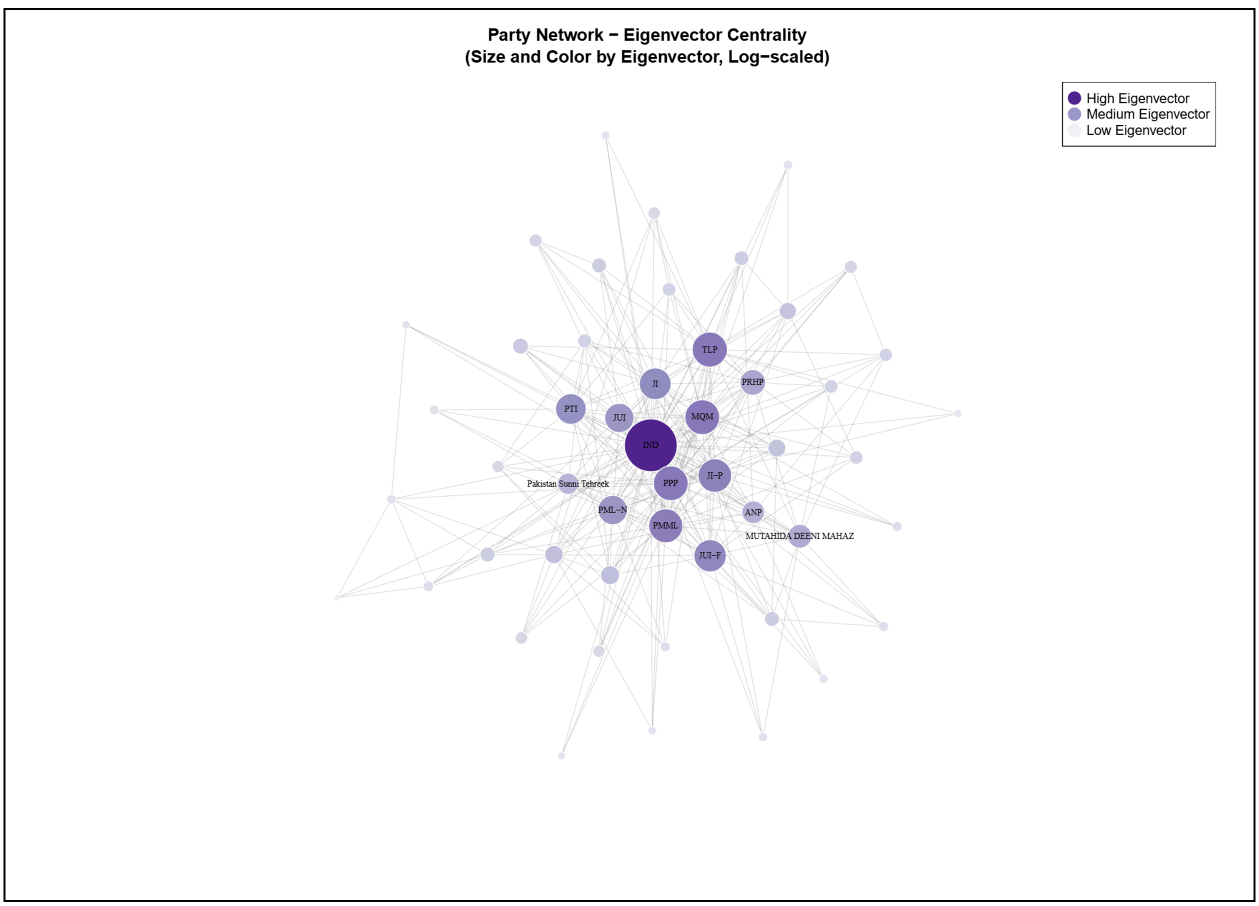

Figure 5: EigenVector Centrality

Eigenvector centrality shows independent candidates as the most central node, which is indicated by the intensity of purple colour and largest in size. In terms of the Eigenvector, it highlights how IND has a systemic influence. Its connections are with other important influential parties. Thus, it shows the important role of independent candidates in the party switching network. Figure 5 also highlights how it is connected with secondary hubs such as PPP, MQM, and TLP. A candidate switching from a peripheral party to the IND or other parties with a high eigen vector is seen as a strategic move in the political system. It highlights that party switching is mostly done to get connected to the most powerful parties, in order to increase political capital and align with the influential political bloc. In this way, it shows over three periods, the pattern of loyalty has been based on access to power and influence.

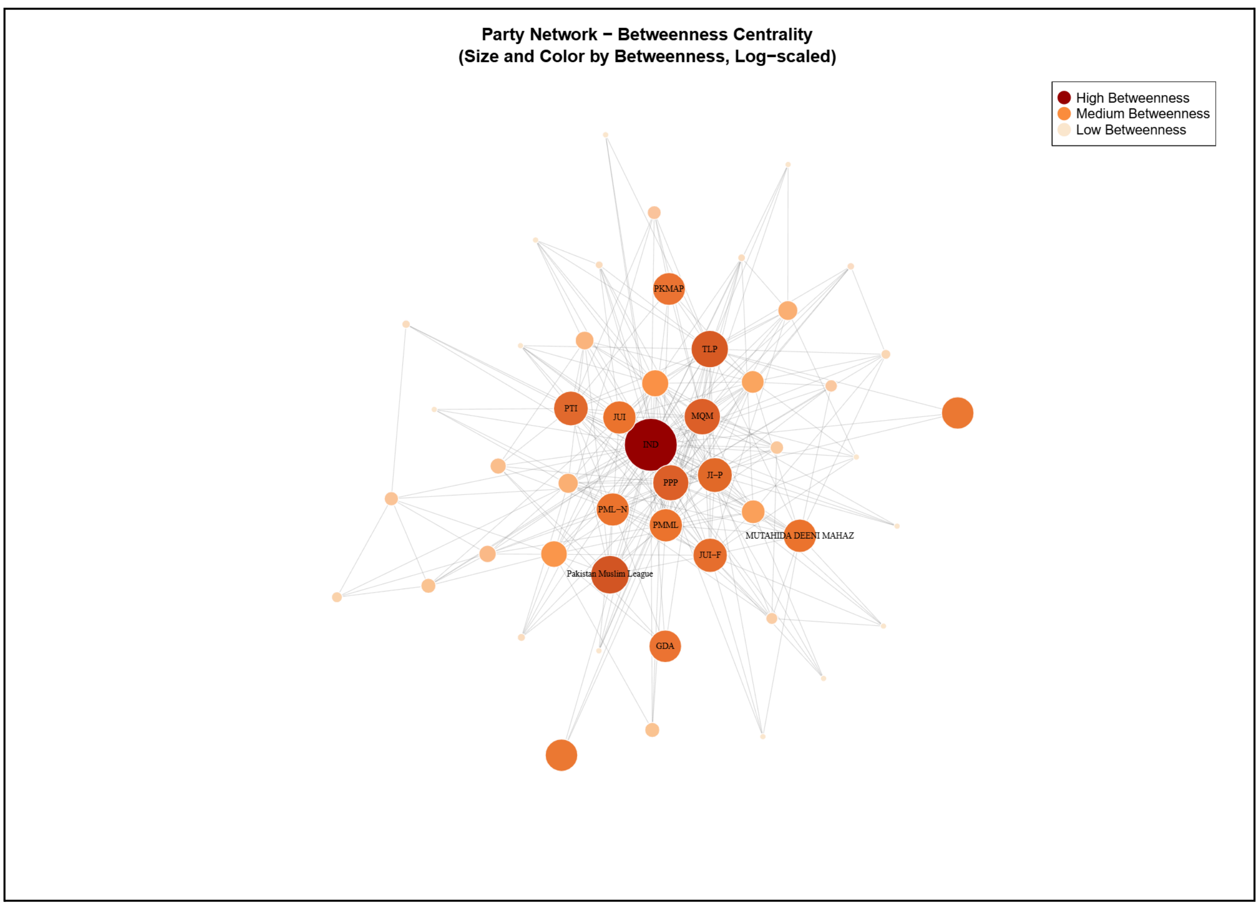

Figure 6: Betweenness Centrality

Betweenness centrality helps to identify the parties/political affiliations that act as a bridge or brokerage of power connecting two otherwise disconnected sections within the network. Here, IND is the main bridge through which candidates mostly switch, thus presenting it as an important neutral zone. It is indicated by the intensity of the orange colour and the largest size. It acts as an important intermediary for candidates to reach their desired parties and connects them to otherwise disconnected political factions. As a result, it has a lot of structural power. Pakistan Muslim League PML has the second-highest betweenness centrality, despite having a low degree, other than core parties, so it also acts as a broker over the period of three electoral cycles. It shows how it is central in terms of its position, which means it connects otherwise disconnected parties, such as connecting regional opposition parties, such as GDA, to religious parties, who are not directly connected. Few parties act as bridge/brokers, so this explains the systemic instability in Pakistani politics, where the coherence of large political blocs relies on the stability of a few broker parties through candidate switching.

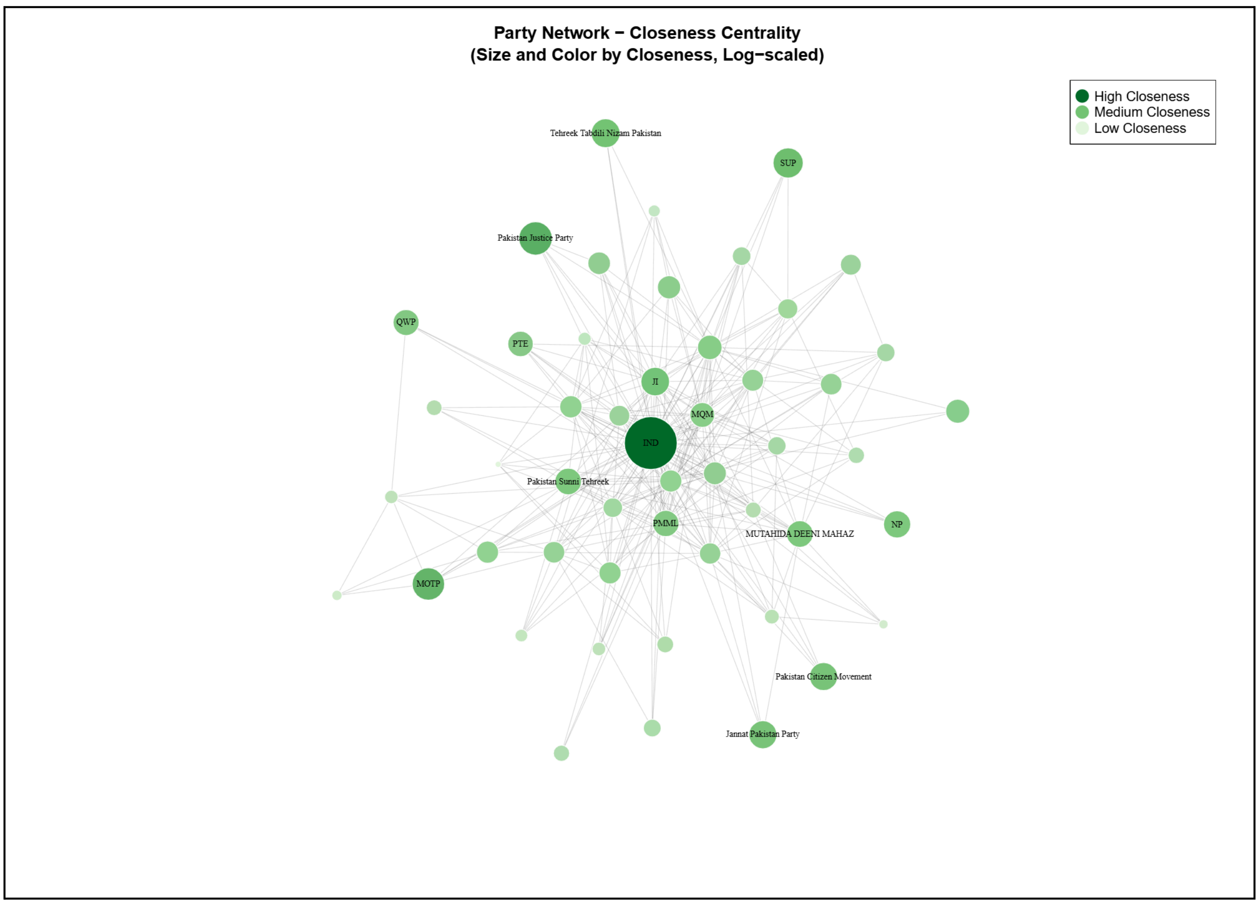

Figure 7: Closeness Centrality

According to closeness centrality, IND (0.55) is closest, and then followed by Pakistan Justice Party and MOTP. This is shown by the intensity of green colour and the large size of the node. Closeness centrality measures how quickly information (in this case flow of candidates) can reach all other nodes in the network starting from a given node. Over the three electoral periods, IND has been the most central, indicating how it acts as a common source between established political parties. Candidates frequently move to or from an independent platform. This highlights how there is high volatility, fragmentation, and low party loyalty. Pakistan Justice Party and MOTP are not as central but still play a role in transmitters or receivers of candidates. They are part of structure in defining how parties interact through candidate movement.

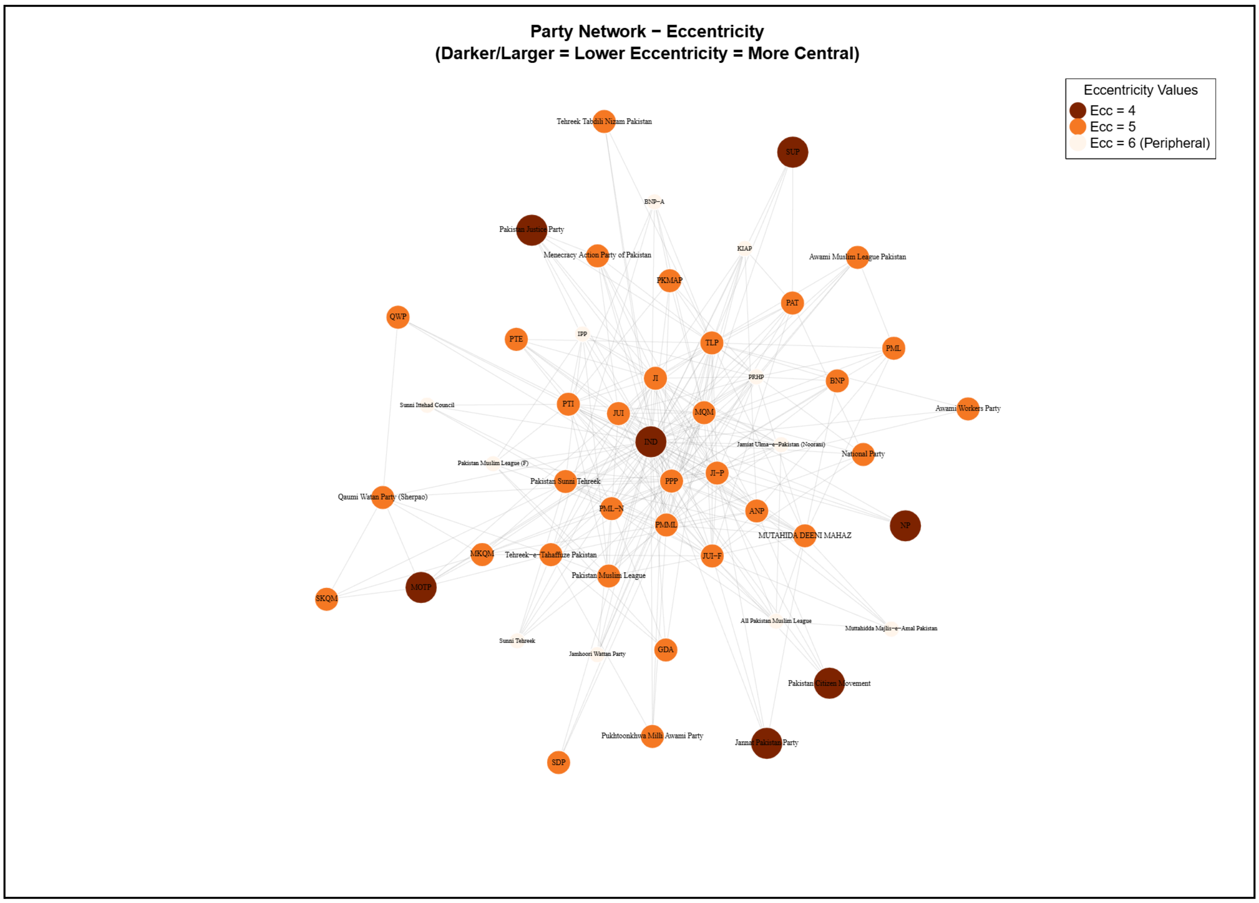

Figure 8: Eccentricity Centrality

Eccentricity of a node is the greatest distance between that node and any other node in the network. A low eccentricity is when the party is close to the furthest party. Here in this network, many parties had 0 eccentricity, which means they have not been involved in the process of where candidates switched from one party to another, not receiving or losing candidates. This shows a more politically fragmented structure where parties do not interact with others and are isolated in terms of candidate switching and reach. In terms of low eccentricity, IND has a low eccentricity of 4 in the network, which shows it's a structural centre where it can quickly reach every other party through a small number of intermediary switches. The above graph shows only low eccentricity, which starts from 4 to a high of 6, as shown from darker shades to lighter shades. The parties with 0 are removed as they are completely disconnected from the network.

Community Structures:

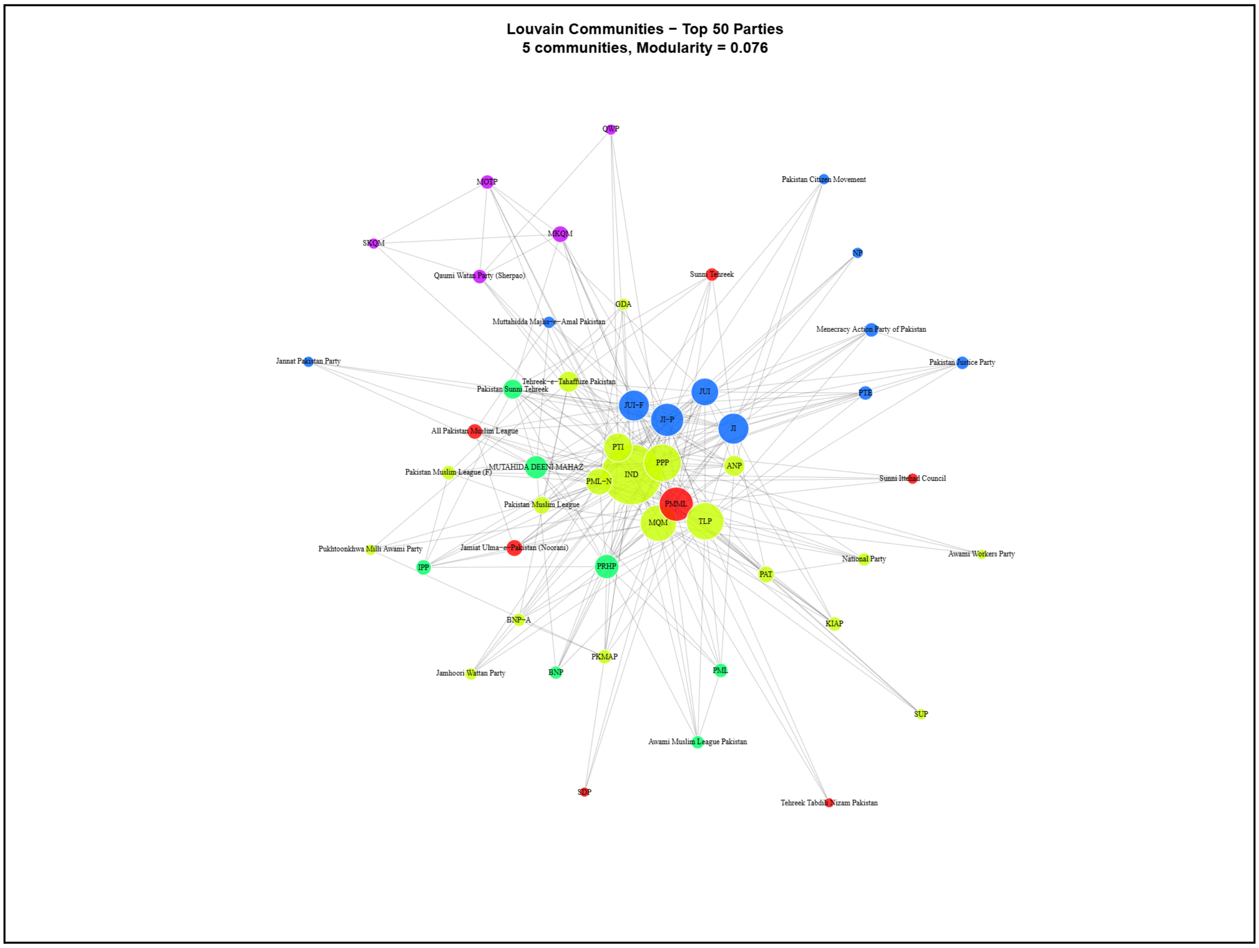

Figure 9: Louvain Communities

The Louvain algorithm highlights low modularity, which shows that community structure is weak, highlighting a highly fluid network where boundaries between blocs can be easily crossed. There is one massive central community which has 54 parties, including IND, PML-N, PPP, PTI, and many more. This community reflects a high rate of candidate flow by moving beyond regional and ideological boundaries. This shows how the candidate switches between parties is non-ideological and mostly driven by volatility. In this community, there is the presence of all major national parties and especially the IND status, which shows it is a major aspect of political instability.

Followed by this community, the second largest community that is more regional and conservative. In this community, there are 24 parties, in which there are religious parties like JI, JUI-F, MMA, MUTAHIDA DEENI MAHAZ, and regional parties such as NP. This bloc is defined by a higher candidate flow that has specific ideological or regional underlying or competition rather than the mainstream power dynamics of the core political structure.

Overall, this community structure reveals how parties that are most crucial for forming governments are tightly knit in a single fluid community and mostly around the IND intermediary, while the rest of the network is defined distinctly but at the same time is connected to conservative/regional blocs. The rest of the vast network is a fragmented periphery made up of many small disconnected parties.

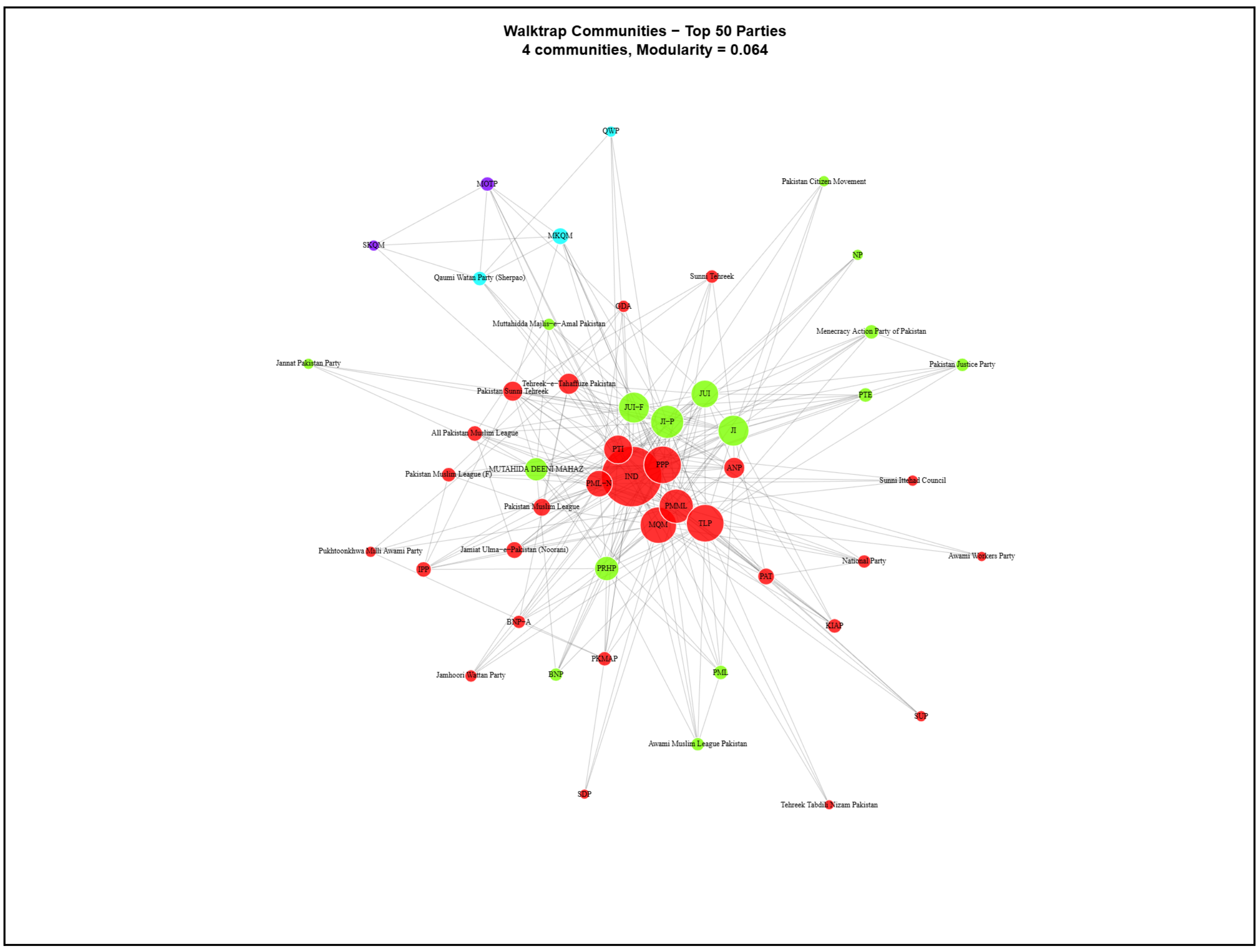

Figure 10: Walktrap Communities

The Walktrap algorithm, which is designed to detect communities by stimulating random walks, identified a highly centralised structure with a low modularity value of ~ 0.064. This highlights that there are weakly separated political blocs in the network. The largest community (community 2) comprises 27 parties including IND, PML-N, PPP, PTI, MQM, PML and many more. This is a more tightly knit community in a similar version as Louvain's. This represents an active and more power sharing dynamics where candidates switch between these major parties and independent status mostly frequently. This shows that political structure is highly fluid and is not constrained by party boundaries.

Similarly, the second largest community (community 5) comprises 12 parties, which is a relatively cohesive religious/ideological bloc that contains major religious parties like JI, JUI-F, and MMA. This indicates that the candidate movement in this space is more based on conservative underlying ideologies. Apart from this, several very small communities show fragmentation within Pakistan's party and political structure, where many parties are isolated from the core in terms of candidate mobility and reach.

Overall, both community detection algorithms show similar structural patterns, showing how over three periods of the electoral cycle, 2008, 2013, and 2024, the political system reflects volatility, and there is wide fragmentation. Both identify how there is a major community bloc which includes all major national parties and the critical role of the IND status. One pattern is that candidate movement goes beyond ideological issues, based on more access to power, incentives and likelihood of being elected, which basically shows political loyalty. On the other hand, there exist smaller, more ideological clusters. However, low modularity in both shows that even within these bloc boundaries are weak and there is candidate flow. Many small disconnected parties around these single cores show how there is instability and fragmentation rather than very strong institutionalisation.